Machine Learning for Detecting Collusion and Capacity Withholding in Wholesale Electricity Markets

Jeremy Proz[‡], Martin Huber[‡], Ivan Marin[†]

This version: April 22, 2026

Abstract

Collusion and capacity withholding in electricity wholesale markets are important mechanisms of market manipulation. This study applies a refined machine learning–based cartel-detection algorithm to two cartel cases in the Italian electricity market and evaluates its out-of-sample performance. Specifically, we consider an ensemble machine learning method that uses statistical screens constructed from offer-price distributions in balancing markets as predictors for the incidence of collusion among electricity providers in specific regions. We also propose novel screens capturing the capacity-withholding behavior of electricity providers and find that including such screens-derived from the day-ahead spot market rather than the balancing market-can improve cartel detection. When the screens are computed from all offers submitted in a market session, both the classic, balancing-market screens and the novel day-ahead screens correctly classify almost all sessions in our data as collusive or competitive, with the combination of both achieving the highest average performance. When the classic screens are computed separately for subsets of participants within a session - an approach tailored to detecting incomplete cartels as some of the bidders in collusive sessions may not participate in the cartel - the performance of our machine learning approach remains similarly high, with correct-classification rates close to 100%. Overall, the study provides a proof of concept for the strong potential of supervised machine learning techniques to detect and dismantle cartels in electricity markets.

[‡]University of Fribourg, Department of Economics, Chair of Applied Econometrics and Policy Evaluation, Bd. de Pérolles 90, 1700 Fribourg, Switzerland. E-mail: jereproz@hotmail.com, martin.huber@unifr.ch

[†]University of Bourgogne, Institut AgroSup, CESAER, Bd. Dr. Petit Jean 26, 21000 Dijon, France. E-mail: ivan.marin@inrae.fr

1 — Introduction

Electricity wholesale in liberalized markets has a history of price manipulation concealed behind market dependencies and tight regulations. The most famous cases include the 2000-2001 California energy crisis (Wolak, 2003) and the 2008 fragmentation of German electricity giant E.ON (Duso et al., 2020) and were eye-opening for understanding electricity wholesale's potential for abuse. Specifically, producers engaged in "capacity withholding" involving many production units where the withdrawal of large quantities of electricity from the day-ahead electricity market was used to artificially inflate the demand on other more profitable markets such as the market for balancing services. Despite economists recently identifying ex-post proof of capacity withholding (see Fogelberg and Lazarczyk, 2019; Bergler et al., 2017; Durmaz et al., 2024) and mounting concerns about unexplained energy price spikes (see Kwoka and Sabodash, 2011; Liu et al., 2024), investigations into suspected illicit activities on wholesale electricity markets often lack a data-driven approach.

The recent literature on data-based cartel detection has developed methods to identify collusion in markets or tenders based on combining behavioral screens - statistics constructed from the price or bid distribution of suppliers - with supervised machine learning for predicting collusion and applied it for instance to the construction sector and consumer goods industry (see Imhof et al., 2018; Huber and Imhof, 2019a; Silveira et al., 2022; Huber et al., 2022; Wallimann and Sticher, 2023; Aaltio et al., 2025; Rodríguez et al., 2022; Aaltio et al., 2025). However, such existing methods cannot account for a key feature of electricity markets that may be valuable for predicting collusion. Specifically, cartel members in wholesale electricity markets may engage in coordinated capacity (or supply) withholding, whereby they strategically reduce the effective availability of capacity in the day-ahead market. This can take the form of physical withholding, i.e. physically reducing the amount of generation capacity offered, or economic withholding, i.e. offering capacity at inflated prices such that it becomes economically unavailable to consumers. In their literature review, Brown (2018) highlight the lack of behavioral screens tailored to the specific features of electricity markets as an important research gap in cartel detection. As a methodological contribution, this study addresses this issue by proposing novel screens for (economic or physical) capacity withholding behavior involving multiple units in the day-ahead market, and by combining them with previously suggested behavioral screens for the balancing market, where producers submit price bids. We demonstrate the method's ability to capture collusive bidding behavior by applying it to two dismantled cases of capacity withholding in Italy and comparing the out-of-sample performance of machine learning in classifying collusive and competitive markets when applying only the previously suggested behavioral screens to the balancing market, only our newly suggested screens to the day-ahead market, and the combined screens. We find that when computing the screens from all offers submitted in a market session, both the classic screens for the balancing market and the new screens for the day-ahead market correctly classify

nearly 100% of the sessions in our data as collusive or competitive. Furthermore, combining both types of screens yields the highest average performance.

We also compute the classic screens separately for all possible subsets of three or four participants within a balancing-market session. The rationale is that - even in collusive sessions - not all producers must be part of the cartel, potentially diluting the cartel's price signal. Focusing on subsets and appropriately aggregating the resulting screens (through summary statistics like the minimum, mean or maximum) may therefore mitigate this dilution when cartels are incomplete; see the discussion in Wallimann and Sticher (2023). When applying this approach, the accuracy (i.e., correct classification rate) of our machine-learning method remains close to 100%. While these results are encouraging, we acknowledge that our setup may be prone to overfitting, as the method is trained and tested on a relatively small number of cartel cases limited in both temporal and geographic scope. This may restrict the external validity of the findings and for this reason, the exercise should be viewed primarily as a proof of concept demonstrating the feasibility of machine learning-based cartel detection. Practical application by regulators would require training and testing on a broader and more diverse dataset with multiple regions and time periods.

As an empirical contribution, our study appears to be the first applying machine learning based on screens to capacity withholding cartels, using public data on offers from the Italian market operator "Gestore dei Servizi Energetici" (GME). More specifically, the two collusive cases underlying our analysis occurred in the Italian electricity market for balancing services (MSD or "Mercato dei Servizi di Dispacciamento"), where capacity withholding in the day-ahead market (MGP or "Mercato del Giorno Prima") increased the reliance on the MSD. The first abuse case concerns energy providers in 2010 in the Campania region in Southern Italy who changed their bidding behavior on the day-ahead energy market to apply a bid rotation scheme in the MSD. Although the cartel went unnoticed, it was subsequently reported by one of the participants, who were all convicted (AGCM, 2012). The second is a case of suspected capacity withholding in 2016 in the Brindisi region of Italy to increase MSD prices. Only one of the providers conceded and was ordered to implement preventive measures under intensive scrutiny while another was not deemed to have a strong enough position on the market to be considered liable. The manipulation was nonetheless estimated to have contributed to cost increases of €320 million in energy production (AGCM, 2017a,b).

Our paper contributes to a growing number of studies using statistical screens for cartel detection. The original literature distinguishes between structural and behavioral screens: the former identifies market structures conducive to collusion, while the latter aims to identify non-competitive behavior, for instance through coordinating or fixing prices and bids (Porter and Zona, 1993; Porter, 2005; Bajari and Ye, 2003; González and Perdiguero, 2011; Harrington, 2005). More recently, behavioral screens have seen another surge in interest when used in conjunction with predictive statistical or supervised machine learning methods, including logistic regression (Silveira et al., 2022), Least Absolute Shrinkage and Selection Operator (or

LASSO, used in combination with other methods), regression trees (Huber and Imhof, 2019a; Huber et al., 2022; Wallimann and Sticher, 2023), random forests (Rodríguez et al., 2022) and Support Vector Machines (SVM) (Wallimann et al., 2022), among others. This study uses an ensemble method that combines multiple machine learning methods for predicting collusion based on screens, following Huber and Imhof (2019a) and Huber et al. (2022). More specifically, our ensemble method consists of a weighted average of five models: random forest, bagged decision trees, LASSO regression, SVM and logistic regression.

We note that Italian wholesale electricity data has also been analyzed by Graf et al. (2020), who examine how physical grid constraints between the day-ahead and subsequent re-dispatch markets shape bidding incentives. They show that the probability of being instructed to increase or decrease production in the re-dispatch market affects firms' day-ahead offer markups, making re-dispatch a lucrative outside option and challenging the view of the two markets as frictionlessly linked. Our study differs from this perspective in two respects. First, while Graf et al. (2020) examine incentive channels between day-ahead and re-dispatch markets, we focus on detecting collusion and capacity withholding directly in the day-ahead market. Second, rather than modeling markups, we consider machine learning-based cartel detection that leverages statistical screens from offer price distributions as predictors. Together, the two approaches provide complementary insights: Graf et al. (2020) identify how institutional design features of the Italian market create opportunities for strategic bidding, while our work demonstrates how detection algorithms can be used to identify instances of collusion in practice.

The remainder of this paper is organized as follows. Section 2 introduces electricity wholesale markets and cartel behavior, and discusses the two Italian cases in further detail. The third section describes the data, the screens, and the machine learning approach. The fourth section discusses the empirical results, in particular the out-of-sample performance in predicting collusive and competitive price offers. The fifth section concludes.

2 — Electricity Markets and Cases

2.1. Institutional Background

According to the OECD (2025), markets with specific structural characteristics are prone to exhibiting non-competitive behavior. Among the main market features is the small number of competitors, complex barriers to entry, predictable demand with limited substitutes, combined with slow technological change and repetitive bidding processes, which create conditions conducive to collusive behavior. In particular, wholesale electricity markets tend to exhibit most of these features (OECD, 2022a). As a consequence, competition authorities have increased their scrutiny through investigations. Specifically, such attention has intensified in a context where market conditions change due to supply shocks or the implementation of algorithmic pricing tools that may strengthen tacit collusion (OECD, 2023). In order to understand how the

structural conditions of wholesale electricity markets promote collusive incentives, we outline the key trading mechanisms that govern the electricity wholesale market.

Unlike other commodities, electricity is delivered through a geographically fixed grid and cannot be stored at scale. Historically, transmission grids were operated by vertically integrated producers. However, the liberalization of European markets between 1996 and 1998 introduced wholesale markets across the European Union (EU) (Parliament, 2026). Such markets are often characterized by separate public authorities that carry out distinct roles in transmission grid maintenance and trading activities. In Italy, the first role is performed by Terna, while the second is carried out by the Electricity Service Administrator (Gestore dei Servizi Energetici, GME).

Additionally, the energy markets involve a vertical structure of producers that supply end-users. Upstream producers sell electricity primarily to retailers in the *wholesale market*. In turn, the downstream firms supply the end-users in the *retail market*. However, several challenges must be overcome to ensure energy supply. First, the physical limits of the transmission network constrain electricity trading. Second, energy flows are also subject to weather conditions, particularly in a context of large renewable energy provision. As a consequence, the TSO manages flows to ensure energy stability. As a result, transmission bottlenecks create geographically bounded submarkets in which a few participants can exert disproportionate influence.

To facilitate trading under these constraints, wholesale electricity markets are organized into standardized products differentiated by the time remaining before delivery. Characterized by its high trading volume, the Day-Ahead Market (Mercato del Giorno Prima, MGP) allows participants to submit hourly day-ahead bids specifying the quantity of electricity they are willing to supply and the corresponding price (in €/MWh). Under perfect competition, these bids would nearly represent their production costs. Following a *pay-as-clear* mechanism, production units are uniformly paid according to a clearing price for each hour, which equals the price at which supply meets demand. This system is designed to promote economic efficiency and transparency (OECD, 2022a). However, producers often rely on demand forecasts and assessments of competitor behavior to prepare bids that exceed breakeven levels.

Congestion and system imbalances that arise after the MGP are addressed in a sequence of progressively shorter-term markets. Among these, the Ancillary Services Market (Mercato dei Servizi di Dispacciamento, MSD) plays a critical role in balancing real-time fluctuations. In principle, highly flexible generation units capable of managing production adjustments are invited to participate in the market. However, current legislation requires all highly flexible generators to participate. The MSD operates within the 24 hours prior to delivery and includes products such as start-up prices, incremental and decremental adjustments, and shut-down prices. Because the units participating in this market are characterized by their high operating costs, MSD electricity prices are typically more expensive than MGP electricity. Unlike the day-ahead market, the MSD uses a *pay-as-bid* system in which each accepted offer is paid at

its own bid price. Although this market design can lead to lower overall expenditure, it also creates incentives for strategic bidding at prices above production costs (OECD, 2022b, p. 14).

In particular, Terna is the only counter-party in the MSD. Terna selects bids according to an optimization criterion aimed at resolving congestion issues and preventing grid overloads. Until 2020, the Italian transmission network was divided into six price zones and four hub areas[1]. Since energy flows between zones depend on constrained high-voltage lines, Terna ensures balance both within and across zones while respecting transmission limits.

In summary, the MSD represents a smaller yet substantially more expensive segment of the wholesale market. Its geographically segmented and time-sensitive design under a *pay-as-bid* structure creates conditions in which a limited number of producers may exert disproportionate influence over market outcomes, reinforcing many of the market's structural vulnerabilities.

2.2. Collusion and Market Abuse

A recent, perhaps pessimistic, estimate suggests that macroeconomic welfare would be 2% higher in a collusion-free utopia (Moreau and Panon, 2022). Another indicator of the large costs of collusion is that the European Regulation on Wholesale Market Integrity and Transparency (REMIT) has resulted in fines totaling €233.5 million since its entry into force in 2015, under REMIT Article 5, for convicted cartels and market manipulators (ACER, 2026). In public procurements such as the MSD, these additional costs are borne by Terna and are therefore directly passed on to the end consumer through their electricity bill.

Firms may refrain from competitive behavior by cooperating with other participants to raise prices, thus forming a cartel. Complete cartels can exert strong control over entrants and prices, while incomplete cartels continue to compete with non-members (McAfee and McMillan, 1992). When working together in a tendering process (known as bid rigging), cartels may use tactics such as cover bidding or a rotation scheme, where bids are structured to ensure one participant wins. Moreover, once a cartel or market abuse is suspected or exposed, participants are often known to revert to non-collusive behavior (Porter, 2005; Wallimann et al., 2022). It has therefore been argued that the threat of discovery is often an effective deterrent (Moreau and Panon, 2022). Participants may also agree to refrain from bidding or competing in certain markets or geographical areas (OECD, 2009).

Specifically, in the electricity wholesale market, refraining from competing in a market session to manipulate prices is a common practice known as *capacity withholding*. This can occur even without collusion if a single participant has sufficient influence over the market. Capacity withholding is specifically addressed by REMIT but is notoriously difficult to prove, as empirical evidence linking it to market abuse is often lacking. REMIT distinguishes between economic and physical capacity withholding. The former occurs when a producer offers its

[1]Unlike normal congestion zones, the hub areas are characterized by a smaller number of participants.

generation capacity at very high prices to avoid being selected, while the latter involves simply not offering the generation capacity at all (ACER, 2024).

2.3. Cases

Italian wholesale electricity markets have been the site of several market abuse cases. In contrast to other European markets, however, the investigation of such cases in Italy is facilitated by the market operator's open data policy. Specifically, wholesale bids and offers from power producers are regularly published, including detailed information on price, quantity, production/consumption units, and participant data. Additionally, accepted and rejected bids, along with their prices, are also made publicly available.

The first case explored in this study involves a bid rotation scheme in the Campania region in 2010, where four units from three separate owners colluded. The Campania region is part of the larger CSUD (Central-Southern Italy) congestion zone, covering the regions of Lazio, Abruzzo, Campania, and Umbria. Between May and October, these producers were the only functioning units on Sundays and holidays in the local "Campania cluster" of production units. It is important to note that this cluster was defined for the specific case and was not a known system weakness (e.g., due to congestion-related problems) at the time. Additionally, by simultaneously and strategically placing high prices on the MGP, the operators of these units realized that at least one offer had to be accepted on the MSD for switching-on (so-called unit startup) products if no offers were accepted on the MGP. During these months, the units intentionally lost MGP offers and took turns meeting the resulting MSD demand, thus delivering a similar output at a higher price than the MGP price. While the average MSD price exceeded the MGP price by two and a half times, GME did not initially suspect a cartel, as MSD units are generally more costly than MGP units. It was only after an anonymous tip that an investigation was launched, leading to the discovery of the cartel, primarily through evidence of secret communications (AGCM, 2012). While this cluster was narrowly defined in the legal case, making it a complete cartel in this region at that time, it evaded GME's detection by operating within a seemingly competitive market.

The second case concerns a similar situation in 2016 in the Brindisi region involving four units from two owners, hereafter referred to as the Brindisi or 2016 case. The Brindisi region constitutes a congestion zone of the same name (noted as BRNN), which, while much smaller than the CSUD area, experiences similar congestion constraints, albeit with fewer participants. Unlike the previous case, these units did not exhibit advanced coordination. Instead, the largest of the two owners (operating three of the four collusive units) initially withheld production on the MGP unilaterally to inflate MSD prices. Another operator subsequently joined, thereby holding a majority influence on the local market. By restricting competition and increasing the necessity for two units to be accepted in the MSD for unit startup, the two participants shared unusually high profits, as the costs of balancing services for Terna were €320 million higher than the previous year. Although the collusion period cannot be precisely defined, it

reached its peak when both parties began systematically withholding on April 21st and ended after the legal investigation began on June 15th (AGCM, 2017a,b). In the absence of explicit communication, the two partners in this tacit collusion were not charged with collusion but rather based on individual behavior, although it is clear that both benefited from the situation, thus constituting a case of tacit collusion.

3 — Data and Screens

3.1. Data Processing and Market Definition

This study uses public tender data from the energy market platform of the Electricity Service Administrator (Gestore dei Servizi Energetici, GME). We downloaded all tenders from the Day-Ahead Market (Mercato del Giorno Prima, MGP) and the Ancillary Services Market (Mercato dei Servizi di Dispacciamento, MSD) for 2010 and 2016. Given the relevant markets associated with these cases, we implement a preprocessing and filtering step to select relevant tenders. First, we filter out all replaced and revised offers. Additionally, we exclude offers for the exceptional 25th hour during the daylight savings time change in October. For the MSD data, only offers submitted for unit startup are considered. Collusive periods are defined by each case, while non-collusive tenders are defined as those observed after the end of the cartel period.

Two levels of market definition are proposed based on the availability of MSD data. First, a *narrow* market definition is aligned with the notion of high market power. Second, a broader market definition, namely the *broad* market definition, corresponds to the congestion regulation zones. The first definition is intended for *ex-post* identification purposes, while the second is intended for *ex-ante* identification purposes. As described above, the *narrow* definition for the Campania case includes cartel participants on Sundays and holidays between May and October, whereas the *broad* definition extends the market to all participants in the Central South congestion region (CSUD) over the entire period. For the Brindisi case, the *narrow* definition includes all cartel participants during April to June, while the *broad* definition includes all participants in the Brindisi congestion region (BRNN). Our empirical models use the narrow definitions for complete cartel detection and the broad definitions to test for incomplete cartels (Wallimann et al., 2022).

To prevent model bias, we implement two further processing steps. First, identical bidding behavior across separate tenders is removed so that the models do not place unnecessary emphasis on any specific bidding pattern. Second, we address the limited number of collusive tenders by using *undersampling*, which allows us to independently and randomly sample from the collusive and non-collusive pools of tenders. The first step prevents the model from conflating collusive and competitive behavior. The second step handles the class imbalance common in machine learning applications and is applied uniformly to both cases, except for the narrow market in the Campania 2010 case.

Finally, to test the generalizability of our method, we combine the data from each case into a large pooled dataset for both narrow and broad market definitions. The narrow market is intended to support ongoing investigations with statistical evidence *ex-post*, while the broad market is intended to test whether the method can be applied *ex-ante*.

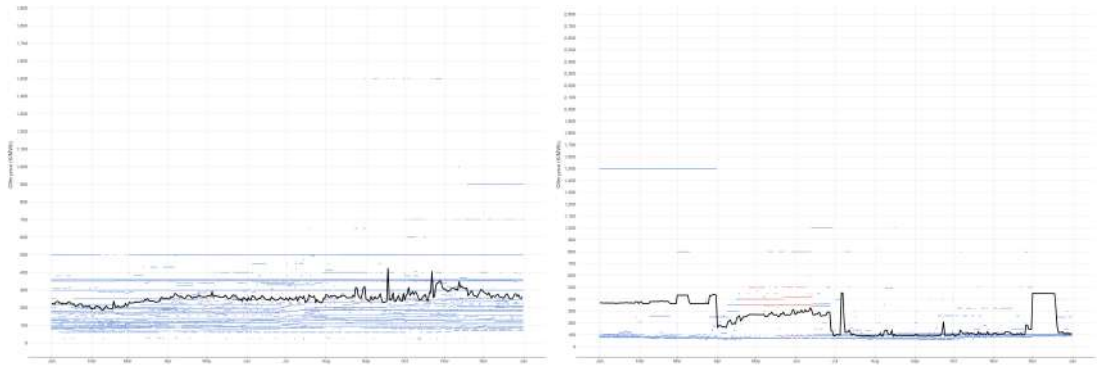
Table 1 summarizes the data used for the models. Despite the large number of available tenders, we retain only tenders from the established collusive period and the non-collusive period. The deduplication of identical bidding behavior had no effect on sample size. Under-sampling, however, effectively reduced a considerable number of tenders. For the Campania case, the narrow definition includes only 54 tenders. For all other market definitions in both the Campania and Brindisi cases, we maintain a balance of 50 percent between competitive and collusive tenders. Since the combined datasets include both cases in a single framework, these totals correspond to simple additions. At first glance, both the Brindisi and Campania cartels sustained similar bidding rates, with 7 and 6 offers per tender on average, respectively. The broad definitions, however, reveal a considerable difference in the average number of offers between the two congestion regions.

Table 1: Processed MSD tenders

Case	Market definition	Preprocessed Tenders	Prop. of Collusive Tenders	Processed Tenders	Avg. num. offers
Campania (2010)	Narrow	1319.0	0.5	54.0	7.0
Campania (2010)	Broad	8759.0	0.1	528.0	30.6
Brindisi (2016)	Narrow	8783.0	0.2	460.0	6.0
Brindisi (2016)	Broad	8783.0	0.2	490.0	8.0
Combined	Combined	10102.0	0.2	514.0	6.1
Combined	Combined	17542.0	0.1	1018.0	19.7

Figure 1 shows submitted offers in the MSD for the Campania and Brindisi cases over 2010 and 2016, respectively. The line represents the average offer price, and the red points represent bids from the collusive units on the respective collusive days. From GME's perspective, it was likely challenging to determine whether Terna was overpaying in the balancing services market. Although average prices were not exceptionally high, an upward pattern is visible during the collusive period.

Figure 1: Offer prices MSD market



(a) Campania (2010) (b) Brindisi (2016)

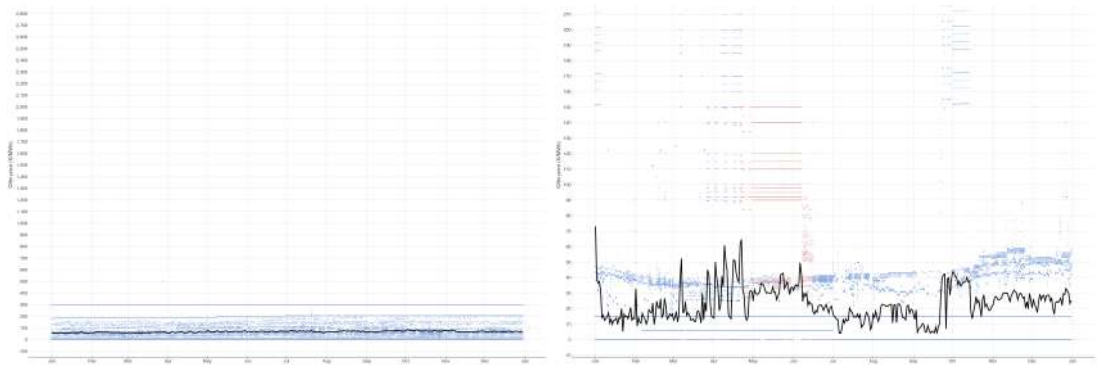
One of the main motivations of the current research is to augment the predictive models with screens derived from the MGP market, which are intended to identify capacity withholding. However, it should be noted that the MGP is characterized by high volatility compared to the MSD, so that adjacent hours exhibit different offer distributions. Since the collusive and non-collusive periods were already defined in the MSD preprocessing step, the MGP data mirrors the sampled days. That is, the *narrow* market is defined by the same collusive days selected for the CSUD and BRNN zones, respectively. Table 2 summarizes the tender data from the MGP market, which is used to construct screens for identifying collusive behavior from the distribution of all MGP offers.

Table 2: Processed MGP tenders

Case	Market definition	Processed Tenders	Avg. num. offers
Campania (2010)	Narrow	54.0	84.8
Campania (2010)	Broad	523.0	87.5
Brindisi (2016)	Narrow	460.0	6.3
Brindisi (2016)	Broad	490.0	6.3
Combined	Combined	514.0	14.5
Combined	Combined	1013.0	48.2

Figure 2 shows observed offers over the sampled period. In the Campania case, the absence of offers during the collusive period—despite the small number of such dates—is consistent with capacity withholding. In contrast, in the Brindisi case, firms were not precluded from submitting bids in the MGP but did so at higher prices. The former case is thus more readily associated with capacity withholding, while the latter makes it more difficult to establish the existence of such an agreement.

Figure 2: Offer prices MGP market



(a) Campania (2010) (b) Brindisi (2016)

3.2. MSD Screens

Following the literature on behavioral screens for detecting collusive behavior, including (Imhof et al., 2018; Huber and Imhof, 2019b), and many others, we use offer prices on the MSD for unit startup in each hourly bidding session to compute 12 different screens representing various features of the offer distribution. The screens are described in Huber et al. (2022) and capture variance, spread, asymmetry, and the overall distributional shape of bids.

A first class of screens concerns the dispersion of offers. Collusion has been found to reduce variation in prices, see for instance the empirical references provided in Huber et al. (2022). A likely reason is that cartel members coordinate bids above the lowest bid that would occur under competition, while avoiding excessively high offers that may attract regulatory scrutiny. To capture this phenomenon, we compute the variance of offer prices as well as the coefficient of variation, defined as the ratio of the standard deviation to the average offer in a market session. We also compute the spread, which measures the relative difference between the highest and lowest offer in a session, i.e., the difference between the maximum bid and the minimum bid divided by the minimum bid. A further dispersion-related measure is the kurtosis, which increases when bids become excessively concentrated around their mean. When cartel members coordinate on similar bid levels - particularly when submitting cover bids - kurtosis tends to increase.

A second class of screens captures asymmetries in the distribution of offers. Collusive bidding often produces unusually large gaps between the lowest and second-lowest offers because the designated winner bids comparably low while other cartel members submit cover bids at coordinated levels. To measure these asymmetries, we compute several variants of the difference between the two lowest bids. The simplest measure is the actual difference between the two lowest prices, $b_{2t} - b_{1t}$, where b_{1t}, b_{2t} denote the lowest and second-lowest offers in session t. In addition, we consider 4 normalized variants of that difference. The first is the percentage difference, given by $(b_{2t} - b_{1t})/b_{1t}$. We also consider the difference normalized by the variation among losing offers, $(b_{2t} - b_{1t})/s_{losing,t}$, where $s_{losing,t}$ is the standard deviation

of losing offers. Further normalized differences compare the gap between the two lowest offers to typical spacing in the price distribution. One measure divides by the mean difference between all adjacent offers in a session, $(b_{2t} - b_{1t})/\{[\sum(b_{i+1,t} - b_{i,t})]/(n_t - 1)\}$, with bids ordered in increasing order. Another measure divides instead by the mean difference between adjacent losing bids, $(b_{2t} - b_{1t})/\{[\sum(b_{i+1,t} - b_{i,t})]/(n_t - 1)\}$. Additionally, we consider the skewness of bids in a session, which captures broader asymmetry in the distribution. Skewness arises, for example, when a low presumed winning bid is combined with a cluster of higher cover bids. A further screen concerns the uniformity of offers. Competitive offers need not be uniformly distributed, but collusive coordination may generate distinctive, irregular patterns. To capture such patterns, we compute the Kolmogorov–Smirnov statistic, (D_t^+, D_t^-) , where $D_t^+ = max_i(x_{it} - \frac{i}{n_t+1})$, $D_t^- = max_i(\frac{i}{n_t+1} - x_{it})$ and x_{it} denotes the standardized bid for rank i. The standardized bid x_{it} is obtained by dividing bid b_{it} by the standard deviation of bids in session t, enabling normalized comparisons of sessions with different price levels. Higher values indicate that the empirical distribution differs more strongly from a uniform distribution, which may reflect collusive structuring of bids. Finally, also the total number of bids is used as a screen.

When applied to wholesale electricity markets, these previously used screens focus on strategic price coordination, since they target submitted offers. They may therefore help detect market abuse in the MSD through coordinated pricing by energy providers, which can affect the asymmetry and variability of offers in market sessions. For instance, variance, coefficient of variation, and spread can indicate coordinated pricing around particular levels. Skewness and the differences between the lowest bids might capture asymmetries that may arise when a designated winning bid is combined with higher cover bids, reflecting collusion in multiunit bidding. The Kolmogorov–Smirnov statistic and kurtosis can detect atypical distributions, such as clustering of offers or unusually concentrated bids, which may result from collusion exploiting sequential clearing incentives in the MGP and MSD markets.

Beyond screens computed on the full set of offers, Wallimann et al. (2022) propose calculating each screen for all possible combinations of three and four offers within a session. This is particularly useful for detecting incomplete rather than complete cartels, where only a subset of suppliers in a session colludes, thereby affecting only the price distribution within that subgroup. By computing summary statistics - such as the maximum, minimum, or median - across these subgroup-based screens within a session, the method may more effectively capture the strategic bidding behavior of incomplete cartels, compared to screens based on the overall price distribution, which also reflects the offers of non-colluding participants. Since both collusion cases in our study involve four colluding units, this method appears highly relevant for our context.

Table 13 in Appendix A shows that the screens differ significantly between collusive and non-collusive sessions in the MSD (when applied to all offers in the respective sessions), based on Mann-Whitney and Kolmogorov-Smirnov tests. This confirms that the features these

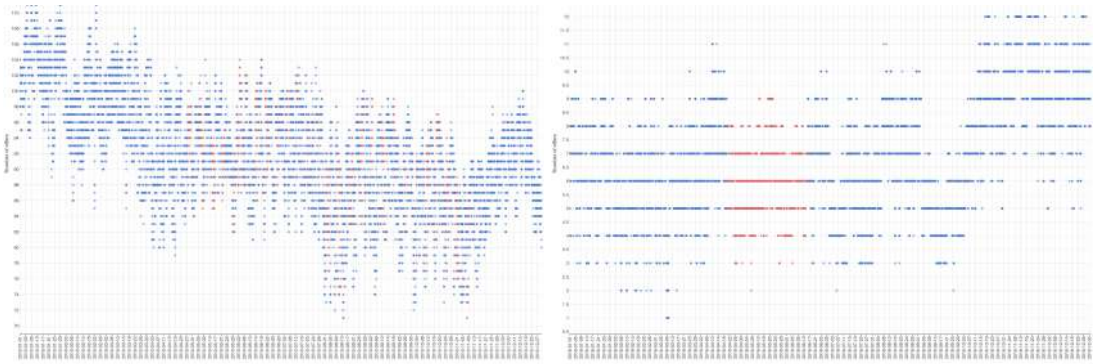
screens capture - reduced variance, increased asymmetry, and atypical spacing between key bids - are systematically linked to collusive conduct.

3.3. Newly Proposed MGP Screens

This section proposes novel screens for collusive behavior in the MGP related to capacity withholding, which differ from the traditionally applied screens in the cartel detection literature. Specifically, we consider four MGP features as screens that exhibit significant differences between collusive and competitive periods.

First, the total number of offers. This screen is analogous to the number-of-bids screen used for the MSD, but applied to MGP data. Unlike the MSD, where qualified units are required to participate, participation in the MGP is voluntary. During capacity withholding events, the number of MGP participants is likely to decrease by at least the number of withholding units. Figure 3 shows the number of offers in the MGP for the Campania and Brindisi cases, respectively, with red points indicating the collusive period.

Figure 3: Number of offers per hour



(a) Campania (2010) (b) Brindisi (2016)

Table 3 shows statistical differences in the distribution of the number of offers between collusive and non-collusive periods, across market definitions and cases. The statistical tests suggest that the difference in the number of bids between collusive and non-collusive periods is statistically significant.

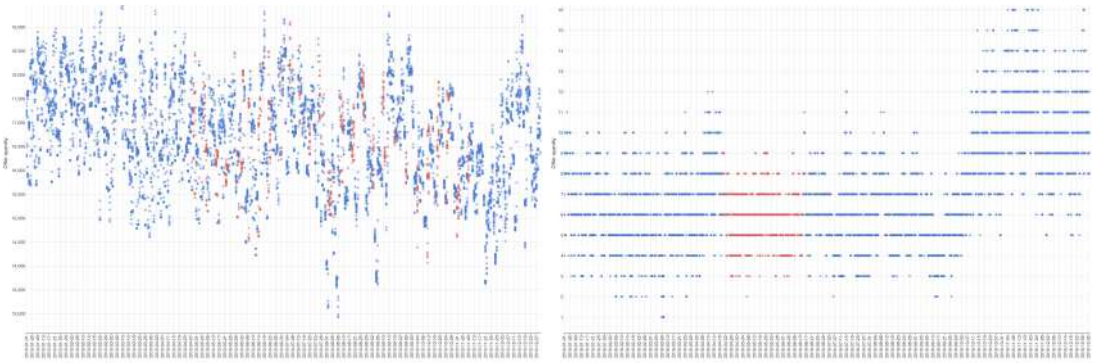
Table 3: Collusive and non-collusive period comparison on MGP offers

Cases	Market	MW stat.	MW pval.	KS stat.	KS pval.	Mean diff.
Campania (2010)	Narrow	287.500	0.678	0.236	0.555	-0.535
Campania (2010)	Broad	20660.000	0.000	0.330	0.000	7.088
Brindisi (2016)	Narrow	18918.500	0.000	0.309	0.000	1.070
Brindisi (2016)	Broad	21788.500	0.000	0.294	0.000	0.959
Combined	Narrow	28636.000	0.010	0.204	0.000	-8.114
Combined	Broad	107128.500	0.000	0.169	0.000	3.650

Notes. MW stands for Mann-Whitney statistical test, while KS stands for Kolmogorov-Smirnov test. The mean difference between non-collusive and collusive period is shown in the last column.

Second, the total offered quantity (expressed in MW/h). This screen is designed to provide insights into the number of participants and the extent of cartel members' influence on market prices. It is derived from the empirical observation that participants can manipulate the market by withholding large quantities.

Figure 4: Offered quantity



(a) Campania (2010) (b) Brindisi (2016)

Statistical tests for the second MGP screen are shown in Table 4. The results from the statistical tests suggest differences in the distribution of offered quantities between periods for both cases, pointing to this screen as a potentially informative additional feature.

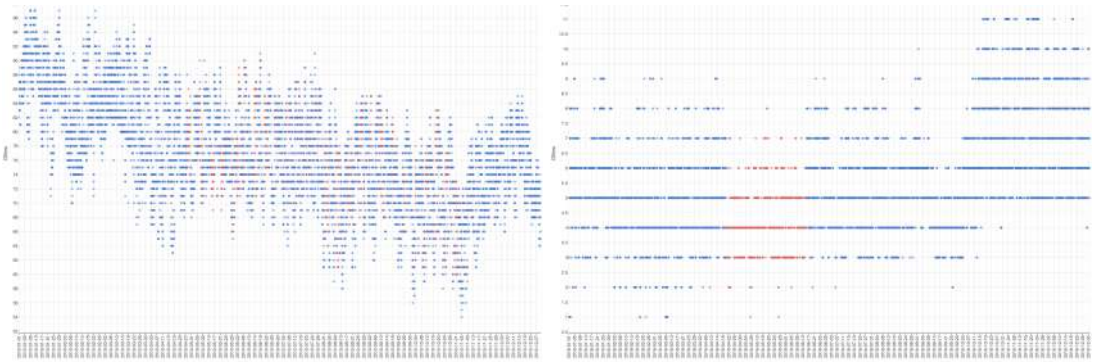
Table 4: Collusive and non-collusive period comparison on MGP offer quantity

Cases	Market	MW stat.	MW pval.	KS stat.	KS pval.	Mean diff.
Campania (2010)	Narrow	430.000	0.001	0.574	0.002	-812.184
Campania (2010)	Broad	24107.000	0.000	0.325	0.000	517.787
Brindisi (2016)	Narrow	35234.000	0.000	0.465	0.000	-235.060
Brindisi (2016)	Broad	37989.000	0.000	0.453	0.000	-165.538
Combined	Narrow	45094.000	0.000	0.449	0.000	-1698.084
Combined	Broad	126776.000	0.749	0.221	0.000	115.179

Notes. MW stands for Mann-Whitney statistical test, while KS stands for Kolmogorov-Smirnov test. The mean difference between non-collusive and collusive period is shown in the last column.

Third, the total number of accepted offers. As is well established in the cartel detection literature, cartel members may submit artificially high prices to avoid winning. This behavior constitutes a form of capacity withholding, and a screen that captures the relationship between winning and losing offers is therefore needed. We use the total number of accepted offers as a third screen. Figure 5 displays the accepted offers in each case.

Figure 5: Accepted offers



(a) Campania (2010) (b) Brindisi (2016)

As with the previous screens, Table 5 summarizes statistical test results on the number of accepted offers by case and market definition. Although the results suggest no significant difference in the distribution for the narrow market in the Campania case, the remaining screens indicate quite different distributions.

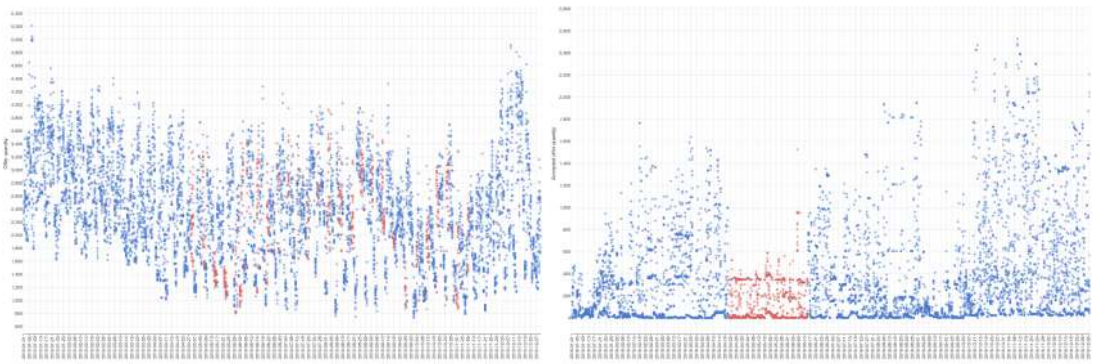
Table 5: Collusive and non-collusive period comparison on MGP accepted offers

Cases	Market	MW stat.	MW pval.	KS stat.	KS pval.	Mean diff.
Campania (2010)	Narrow	262.500	0.943	0.167	0.893	0.559
Campania (2010)	Broad	20496.500	0.000	0.330	0.000	6.081
Brindisi (2016)	Narrow	12615.500	0.000	0.400	0.000	1.613
Brindisi (2016)	Broad	16719.000	0.000	0.339	0.000	1.355
Combined	Narrow	22308.000	0.000	0.294	0.000	-5.741
Combined	Broad	101895.500	0.000	0.169	0.000	3.416

Notes. MW stands for Mann-Whitney statistical test, while KS stands for Kolmogorov-Smirnov test. The mean difference between non-collusive and collusive period is shown in the last column.

Finally, the accepted offered quantity (in MW/h). This screen is the combination of offered quantity and accepted offers. Figure 6 shows the accepted quantities in each case. Table 6 summarizes the statistical test on the screens distribution across periods. Based on the statistical test results, all screens result in a statistically significant difference, except for the Campania case.

Figure 6: Accepted offer quantity



(a) Campania (2010) (b) Brindisi (2016)

Table 6: Collusive and non-collusive period comparison on MGP accepted offer quantity

Cases	Market	MW stat.	MW pval.	KS stat.	KS pval.	Mean diff.
Campania (2010)	Narrow	319.000	0.293	0.274	0.370	-103.926
Campania (2010)	Broad	9665.000	0.000	0.643	0.000	960.897
Brindisi (2016)	Narrow	18558.500	0.000	0.252	0.000	140.225
Brindisi (2016)	Broad	23164.500	0.000	0.216	0.000	97.820
Combined	Narrow	28085.500	0.004	0.196	0.000	13.761
Combined	Broad	96664.500	0.000	0.330	0.000	535.679

Notes. MW stands for Mann-Whitney statistical test, while KS stands for Kolmogorov-Smirnov test. The mean difference between non-collusive and collusive period is shown in the last column.

3.4. Machine Learning Models

Once the offer distribution is represented through screens, detecting cartels on the MSD becomes a binary classification task: a standard machine learning problem that determines whether a distribution should be classified as collusive or non-collusive. Following Huber and Imhof (2019a); Wallimann and Sticher (2023), we use an ensemble method as the machine learner, which involves training multiple machine learning algorithms and predicting based on a weighted average of their outputs. This tends to improve the generalization ability of individual models across different datasets (Zhou, 2012, p. 14). The ensemble consists of five models: random forest (Tin Kam Ho, 1995; Breiman 2001), bagged decision trees (Breiman, 1996), LASSO (Tibshirani, 1996), Support Vector Machines (SVM) (Boser et al., 1992; Cortes and Vapnik, 1995), and logistic regression. Each method, as well as the ensemble approach, yields a probability of collusion, i.e., a value between 0 and 1. A market session is classified as collusive if this probability exceeds 0.5.

To train the machine learning algorithm such that it learns the optimal predictive model for collusion from the data and to assess its out-of-sample performance, we randomly split the data into training samples containing 75% of the observations in the respective dataset and test samples consisting of the remaining 25%. Model parameters are estimated in the training data based on 10-fold cross-validation to select the optimal ensemble method. We iterate the train-test sampling process with k-fold approach in 10 simulations.

It must be noted that our splitting method differs from alternative time series cross-validation techniques such as rolling origin or blocked time splits (Kingphai and Moshfeghi, 2022). Our approach relies entirely on the cross-sectional nature of the screens, which are computed from the distribution of bids within each observation and therefore contain no autoregressive component. Since the ultimate goal is a classification task between collusive and competitive classes, our approach is suited to identify the statistical fingerprint of collusive behavior rather than its temporal dimension. This is the standard approach in the ML literature for cartel detection (Huber et al., 2022; Huber and Imhof, 2019b). Finally, stratified random splitting ensures that both classes are represented in every training and test partition, unlike a temporal split, which would mechanically concentrate collusive and competitive observations in separate time blocks and prevent the learner from observing both classes during training.

The three performance measures considered are the overall share of correctly classified market sessions in the test data (accuracy), the share of correctly classified collusive sessions among all collusive sessions (recall or true positive rate) and the share of correctly classified competitive sessions among all competitive sessions (specificity or true negative rate). We repeat the process of randomly splitting the data into training and test samples, training the ensemble and assessing its out-of-sample performance 10 times, reporting the averages of the respective performance measures over the 10 repetitions.

4 — Results

4.1. Complete and Incomplete Collusion on the MSD

We investigate the out-of-sample performance of our machine learning approach using a restricted dataset that includes only the standard MSD screens. Each model applies a different normalization: the normal specification uses raw screens, while the demeaned and standardized specifications apply the corresponding transformations. Additionally, broad market models include screens computed for all subsets of 3 and 4 bids within a session (see Section 3.2).

The results of complete and incomplete collusion are reported in Table 7, which reports the precision, recall, and specificity of each model specification. For comparison, Appendix D reports performance measures for two baseline models: a SVM (Subsection D.1) and a LASSO (Subsection D.2) model. Despite the optimistic results from a nonlinear model such as the SVM, its performance decreases in the Brindisi case. Meanwhile, the linear model is not capable of surpassing the high performance of our proposed ensemble method. Therefore, we discuss the results from our ensemble method in what follows.

As shown in Table 7, the statistical transformations of the screens did not sharply increase the performance of the models; indeed, only a modest improvement is observed across cases and market definitions. Specifically, the demeaning transformation improves accuracy under the narrow market definition for the Campania case, while both demeaning and standardization improved the accuracy for the narrow market definition in the Brindisi case. However, only the standardization of screens improved the accuracy for the broad market in the Campania case, while both transformations (i.e., demeaning and standardization) improved the accuracy for the Brindisi case in the same market. Both recall and specificity follow almost the same pattern, with close values indicating a high rate of correctly classified collusive sessions (recall) and competitive sessions (specificity).

Table 7: Ensemble method performance for collusive classification

		Narrow			Broad		
Cases	Model	Accuracy	Recall	Specificity	Accuracy	Recall	Specificity
Campania (2010)	Normal	96.429	97.231	95.000	99.545	99.265	99.865
Campania (2010)	Demeaned	97.143	97.231	97.500	99.545	99.569	99.544
Campania (2010)	Std.	95.714	97.231	92.500	99.848	99.851	99.865
Brindisi (2016)	Normal	96.696	98.418	94.905	96.992	98.658	95.289
Brindisi (2016)	Demeaned	96.783	98.418	95.087	97.317	98.992	95.631
Brindisi (2016)	Std.	96.783	98.584	94.905	97.073	98.500	95.652

Notes. The narrow market definition is defined for each case. For the Campania case, the narrow market corresponds to tenders where cartel members participated on Sundays and holidays, whereas for the Brindisi case it involves all cartel members during the collusive period. Broad market definitions are extended to include all market participants in both cases. Additionally, the broad market definition includes screens for extensive subsets of 3 and 4 bidders.

Additional results for the performance measures are placed in Appendix B, Table 14. These include the precision, F1-score, PR-AUC, and average confusion matrix across simulations. All reported models use a 0.5 probability threshold, consistent with ML literature. Consequently, a true positive (TP) corresponds to a correctly predicted collusive observation and a true negative (TN) to a correctly predicted competitive observation. The set of performance metrics confirms the patterns observed in Table 7 and further provides a basis for market monitoring. The precision measure observed across all broad market definitions indicates that every alarm raised by the model would correspond to a genuinely collusive observation, i.e. true positive alarm. Since PR-AUC is consistently around 1, our model provides good separability between collusive and competitive classes when the model has enough information in the training data. Based on the average confusion matrices, the results show that, for instance, in the combined broad market with MGP screens using standardized screens, the model correctly classifies 125 collusive and competitive observations out of the test set, with only 2.4 false positives and 1.6 false negatives on average across the set of simulations.

Based on the results between false positives and false negatives, some implications can be drawn for a wholesale market monitor. The false discovery rate (FDR)-otherwise, the share of flagged as collusive tenders that are indeed competitive-remains below 6 % across all specifications. Specifically, the FDR is below 1% in most of the broad market definitions. Thus, a regulator deploying our model in the combined broad model with the proposed capacity withholding screens would face fewer than 3 false alarms in a test dataset with around 250 observations (TP+TN). The missing rate (otherwise a non detected true positive observations, FNR=FN/(FN+TP)) is also low: for the best-performing combined broad specifications, it oscillates between 1.3% and 2% of collusive tenders not detected. In terms of weekly tenders with 168 sessions (24 tenders on 7 days), the regulator would miss up to 3 sessions if the whole week is collusive. However, given the cases at hand, we view such a scenario as extremely unlikely.

The narrow market results illustrate the cost of limited training data. In the CSUD narrow case, although recall and precision remain high (above 95%), the components of the confusion matrix reveal that the test set contains only approximately 11 observations (TP+FP), of which about 2.8 are competitive. With such small samples, a single misclassification has outsized effects on rates: one false negative shifts the miss rate by nearly one percentage point. The BRNN narrow case, with a larger test set (approximately 58 collusive, TP+FN, and 55 competitive observations, TN+FP), yields more stable estimates, with FDR around 4% and miss rate between 1.2% and 1.5%.

From a policy perspective, these results suggest that the ensemble approach is well suited for deployment as a first-stage monitor tool in incomplete cartel detection, where broad market definitions provide sufficient data for reliable training. The low FDR ensures that regulatory resources are not wasted on false leads, while the high recall minimizes the risk of undetected collusive behavior. For complete cartel detection on narrow markets, the model remains in-

formative but should be complemented with additional qualitative analysis given the limited sample sizes. More broadly, the differential performance between complete and incomplete cartel detection reinforces the view that market-wide screening — capturing the full set of participants in a congestion zone — is more effective than firm-level monitoring.

Table 8 reports learner weights for each case and market definition, revealing notable patterns across screens and their transformations. In the narrow market, weights vary across cases. Although Random Forests and Bagged Decision Trees consistently receive the highest weights, less agreement is observed among the remaining learners: the SVM ranks third for the Campania case, while logistic regression receives a large weight for the Brindisi case. Weights also exhibit sensitivity to screen transformation. Under demeaning, the Campania case favors the LASSO and SVM, while Random Forests and Bagged Decision Trees dominate for the Brindisi case. Under standardization, both cases converge: Random Forests and Bagged Decision Trees receive the highest weights in all specifications. The broad market definition yields a still clearer picture, with these two learners dominating regardless of model specification or screen transformation.

These results indicate that, despite the large number of features used, both Random Forests and Bagged Decision Trees are able to distinguish between collusive and non-collusive tenders. The method therefore demonstrates robustness to large feature sets, even when they include weakly predictive variables that could otherwise degrade ensemble performance.

Table 8: Learner weights for ensemble method

		Narrow					Broad				
Cases	Model	Random Forest	Bagged Decision Trees	LASSO	SVM	Logistic	Random Forest	Bagged Decision Trees	LASSO	SVM	Logistic
Campania (2010)	Normal	0.221	0.221	0.121	0.221	0.215	0.207	0.206	0.183	0.202	0.201
Campania (2010)	Demeaned	0.201	0.201	0.204	0.203	0.191	0.208	0.208	0.175	0.204	0.205
Campania (2010)	Std.	0.204	0.204	0.197	0.199	0.196	0.202	0.202	0.198	0.197	0.201
Brindisi (2016)	Normal	0.242	0.241	0.135	0.143	0.240	0.213	0.214	0.158	0.204	0.212
Brindisi (2016)	Demeaned	0.231	0.231	0.169	0.139	0.230	0.202	0.203	0.200	0.194	0.201
Brindisi (2016)	Std.	0.201	0.201	0.199	0.199	0.199	0.201	0.202	0.201	0.197	0.200

Taken together, these results show that statistical transformations (demeaning and standardization) have only a modest effect on both performance and learner weights. Performance improvements are minor and occasionally negative—as for the Campania case under standardization—while learner weights can shift noticeably, as illustrated by the demeaned screens for the Campania case.

4.2. Addition of MGP Screens

Table 9 reports results for models augmented with MGP screens. Although the accuracy gains are modest, the results suggest that capacity withholding screens carry information beyond the standard MSD screens. Accuracy increases across both cases and market definitions when raw

(normal) screens are added, with the largest gain in the narrow market—an average improvement of 0.5 percentage points. Under the broad market definition, where incomplete cartel information enters through subset-based screens, the MGP screens play a more decisive role when data are scarce. Statistical transformations of the MGP screens, however, offer no consistent benefit: they either reduce accuracy (e.g., Campania case, narrow market, demeaned) or leave the improvement unchanged (e.g., Campania case, broad market, demeaned).

Table 9: Ensemble method performance for collusive classification: Original vs. MGP screens

			Accuracy	
Market	Case	Model	Original	With MGP screens
Campania (2010)	Narrow	Normal	96.429	97.143
Campania (2010)	Narrow	Demeaned	97.143	96.429
Campania (2010)	Narrow	Std.	95.714	97.143
Brindisi (2016)	Narrow	Normal	96.696	97.304
Brindisi (2016)	Narrow	Demeaned	96.783	97.304
Brindisi (2016)	Narrow	Std.	96.783	97.217
Campania (2010)	Broad	Normal	99.545	99.697
Campania (2010)	Broad	Demeaned	99.545	99.545
Campania (2010)	Broad	Std.	99.848	99.848
Brindisi (2016)	Broad	Normal	96.992	97.236
Brindisi (2016)	Broad	Demeaned	97.317	97.642
Brindisi (2016)	Broad	Std.	97.073	97.154

Building on the weight analysis above, Table 10 examines how incorporating MGP screens affects the weights of the two most prominent learners. Somewhat counterintuitively, adding capacity withholding screens tends to reduce the weights of both Random Forests and Bagged Decision Trees. For the Random Forest, this decrease holds for both cases in the narrow market with normal screens, and for the Brindisi case in the broad market. Bagged Decision Trees follow the same pattern in the Campania narrow market, although a marginal weight increase is observed for Brindisi in the narrow market. In the broad market, weights remain unchanged for the Campania case while both learners decline for the Brindisi case. Under statistical transformations, weights are broadly stable across broad markets, with only a marginal increase for the Campania narrow market.

Table 10: Ensemble learner weights: Original vs. MGP screens

			Random Forest		Decision Bagged Trees	
Market	Case	Model	Original	With MGP screens	Original	With MGP screens
Narrow	Campania (2010)	Normal	0.221	0.205	0.221	0.204
Narrow	Campania (2010)	Demeaned	0.201	0.205	0.201	0.204
Narrow	Campania (2010)	Std.	0.204	0.204	0.204	0.202
Narrow	Brindisi (2016)	Normal	0.242	0.241	0.241	0.242
Narrow	Brindisi (2016)	Demeaned	0.231	0.232	0.231	0.232
Narrow	Brindisi (2016)	Std.	0.201	0.201	0.201	0.202
Broad	Campania (2010)	Normal	0.207	0.206	0.206	0.206
Broad	Campania (2010)	Demeaned	0.208	0.208	0.208	0.208
Broad	Campania (2010)	Std.	0.202	0.202	0.202	0.202
Broad	Brindisi (2016)	Normal	0.213	0.207	0.214	0.208
Broad	Brindisi (2016)	Demeaned	0.202	0.202	0.203	0.203
Broad	Brindisi (2016)	Std.	0.201	0.201	0.202	0.202

To further assess the contribution of the new screens, Figures 7–9 in Appendix E present variable importance plots from the Random Forest, which is well suited to this type of analysis. As Figure 7 shows, screens related to the MGP market vary in relevance across cases and market definitions. For the Campania case, the MGP offered quantity is the most relevant predictor, whereas for the Brindisi case the MGP accepted quantity ranks among the top predictors. Additionally, both offered quantity and accepted offers from the MGP appeared as relevant predictors in the combined data. These results contrast with their absence in the broad market definition, where none of the MGP predictors appears as a key variable identified by the Random Forests. This pattern is also observed using statistical transformations (see Figure 8 and 9).

4.3. Combined Results

To assess the generalizability of our results, we combine both cases into a single dataset for each market definition, thereby testing the ensemble's ability to correctly classify sessions using screens drawn from two distinct cartels. Table 11 reports the accuracy measure of our method on the combined dataset. The ensemble learner achieves around 95 and 98 percent accuracy, somewhat below the individual-case results but nonetheless high. The closeness of performance measure between the combined and separate datasets suggests good generalization, with no strong evidence of overfitting.

As in the individual-case analysis, broad market definitions yield higher accuracy than narrow ones. Meanwhile, the inclusion of capacity withholding screens consistently improves performance, reinforcing the relevance of these new screens.

Table 11: Ensemble method performance for collusive classification: combined datasets

Market	Screens	Model	Accuracy		
			Campania (2010)	Brindisi (2016)	Combined
Narrow	Orginal	Normal	96.429	96.696	95.581
Narrow	Orginal	Demeaned	97.143	96.783	95.736
Narrow	Orginal	Std.	95.714	96.783	95.349
Broad	Orginal	Normal	99.545	96.992	97.804
Broad	Orginal	Demeaned	99.545	97.317	97.961
Broad	Orginal	Std.	99.848	97.073	98.392
Narrow	With MGP	Normal	97.143	97.304	96.202
Narrow	With MGP	Demeaned	96.429	97.304	96.202
Narrow	With MGP	Std.	97.143	97.217	96.279
Broad	With MGP	Normal	99.697	97.236	98.118
Broad	With MGP	Demeaned	99.545	97.642	97.961
Broad	With MGP	Std.	99.848	97.154	98.431

Table 12 compares the weights of the most influential learner (Random Forest) with a simpler benchmark (LASSO) across the individual cases and the combined dataset. In the narrow market, pooling the two cases reduces the Random Forest weight under both specifications—with and without MGP screens—while the LASSO weight increases correspondingly. A similar pattern holds in the broad market, where the Random Forest weight declines and the LASSO weight increases marginally.

Table 12: Ensemble learner weights: combined datasets

Market	Screens	Model	Campania (2010)		Brindisi (2016)		Combined	
			Random Forest	LASSO	Random Forest	LASSO	Random Forest	LASSO
Narrow	Original	Normal	0.221	0.121	0.242	0.135	0.240	0.145
Narrow	Original	Demeaned	0.201	0.204	0.231	0.169	0.240	0.145
Narrow	Original	Std.	0.204	0.197	0.201	0.199	0.201	0.199
Narrow	With MGP	Normal	0.205	0.180	0.241	0.136	0.239	0.147
Narrow	With MGP	Demeaned	0.205	0.180	0.232	0.167	0.241	0.139
Narrow	With MGP	Std.	0.204	0.196	0.201	0.200	0.202	0.199
Broad	Original	Normal	0.207	0.183	0.213	0.158	0.210	0.171
Broad	Original	Demeaned	0.208	0.175	0.202	0.200	0.207	0.180
Broad	Original	Std.	0.202	0.198	0.201	0.201	0.202	0.199
Broad	With MGP	Normal	0.206	0.185	0.207	0.179	0.206	0.184
Broad	With MGP	Demeaned	0.208	0.177	0.202	0.199	0.206	0.183
Broad	With MGP	Std.	0.202	0.198	0.201	0.201	0.202	0.199

5 — Conclusion

This study contributes to the growing literature on cartel detection by empirically testing supervised machine learning techniques on real-world cases of market manipulation in the Italian wholesale electricity markets. By combining behavioral statistical screens with an ensemble learning method, it demonstrates the potential of data-driven approaches to uncover collusive behavior among electricity suppliers in market sessions. As a methodological contribution, the paper introduces four novel screens derived from capacity-withholding behavior in the day-ahead market (MGP), specifically tailored to the structural features of wholesale electricity markets and designed to complement existing screens for the balancing market (MSD).

Under complete cartel configurations, where all suppliers in a collusive session are presumed to participate in the cartel, the ensemble method achieves up to approximately 97% accuracy in classifying sessions as collusive or competitive, representing a meaningful improvement over relying solely on MSD screens. Under incomplete cartel configurations, where only a subset of suppliers colludes and screens are computed over all possible subsets of participants, the newly proposed capacity-withholding screens provide no additional performance gains beyond the existing MSD screens, which already achieve near-perfect classification. When both cases are combined into a pooled dataset, the model maintains accuracy of around 95–98%, suggesting good generalization with limited evidence of overfitting.

From a policy perspective, the results carry immediate practical implications. The low false discovery rate (below 6% across all specifications, and below 1% in most broad market definitions) means that regulators could deploy this model as a first-stage monitoring tool with minimal risk of wasting enforcement resources on false leads. The comparably low missing rate (below 2% for the best-performing specifications) further minimizes the risk of undetected collusion. Together, these properties make the approach well suited for deployment as a real-time screening system in electricity market surveillance.

We acknowledge, however, that the strong performance observed may partly reflect overfitting, as the method is trained and tested on a limited number of cartel cases confined to specific regions and time periods. This constrains the external validity of the findings, and the exercise should therefore be viewed primarily as a proof of concept demonstrating the feasibility of machine learning-based cartel detection in electricity markets. Practical deployment by regulators would require training and testing on a broader and more diverse dataset spanning multiple regions, market structures, and time periods. Future research should also explore the integration of additional market signals, such as real-time grid topology or cross-market correlations, and extend the framework to other jurisdictions where similar data transparency exists.

References

Aaltio, A., Buri, R., Jokelainen, A., and Lundberg, J. (2025). Complementary bidding and cartel detection: Evidence from nordic asphalt markets. *International Journal of Industrial Organization*, 98:103129.

ACER (2024). Acer guidance on the application of regulation (eu) no 1227/2011 of the european parliament and of the council of 25 october 2011 on wholesale energy market integrity and transparency.

ACER (2026). Overview of sanction decisions.

AGCM (2012). Repower italia prezzo dispacciamento energia elettrica centro sud, provvedimento n. 23623.

AGCM (2017a). Enel-prezzi servizi di dispacciamento area brindisi, provvedimento n. 26562.

AGCM (2017b). Sorgenia-prezzi servizi di dispacciamento area brindisi, provvedimento n. 26563.

Bajari, P. and Ye, L. (2003). Deciding between competition and collusion. *Review of Economics and statistics*, 85(4):971–989.

Bergler, J., Heim, S., and Hüschelrath, K. (2017). Strategic capacity withholding through failures in the german-austrian electricity market. *Energy Policy*, 102:210–221.

Boser, B. E., Guyon, I. M., and Vapnik, V. N. (1992). A training algorithm for optimal margin classifiers. In *Proceedings of the fifth annual workshop on Computational learning theory*, COLT92, page 144–152. ACM.

Breiman, L. (1996). Bagging predictors. *Machine Learning*, 24(2):123–140.

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1):5–32.

Brown, D. P. (2018). Capacity payment mechanisms and investment incentives in restructured electricity markets. *Energy Economics*, 74:131–142.

Cortes, C. and Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20(3):273–297.

Durmaz, T., Acar, S., and Kızılkaya, S. (2024). Generation failures, strategic withholding, and capacity payments in the turkish electricity market. *Energy Policy*, 184:113897.

Duso, T., Szücs, F., and Böckers, V. (2020). Abuse of dominance and antitrust enforcement in the german electricity market. *Energy Economics*, 92:104936.

Fogelberg, S. and Lazarczyk, E. (2019). Strategic withholding through production failures. *The Energy Journal*, 40(5):247–266.

González, J. L. J. and Perdiguero, J. (2011). Does rigidity of prices hide collusion? *Documents de Treball (IREA)*, (20):1.

Graf, C., Quaglia, F., and Wolak, F. A. (2020). Simplified electricity market models with significant intermittent renewable capacity: Evidence from italy. Technical report, National Bureau of Economic Research.

Harrington, J. E. (2005). Detecting cartels. Technical report, Working paper.

Huber, M. and Imhof, D. (2019a). Machine learning with screens for detecting bid-rigging cartels. *International Journal of Industrial Organization*, 65:277–301.

Huber, M. and Imhof, D. (2019b). Machine learning with screens for detecting bid-rigging cartels. *International Journal of Industrial Organization*, 65:277–301.

Huber, M., Imhof, D., and Ishii, R. (2022). Transnational machine learning with screens for flagging bid-rigging cartels. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 185(3):1074–1114.

Imhof, D., Karagök, Y., and Rutz, S. (2018). Screening for bid rigging—does it work? *Journal of Competition Law & Economics*, 14(2):235–261.

Kingphai, K. and Moshfeghi, Y. (2022). On time series cross-validation for deep learning classification model of mental workload levels based on eeg signals. In *International Conference on Machine Learning, Optimization, and Data Science*, pages 402–416. Springer.

Kwoka, J. and Sabodash, V. (2011). Price spikes in energy markets:"business by usual methods" or strategic withholding? *Review of Industrial Organization*, 38(3):285–310.

Liu, C., Shao, Z., Jiao, J., and Yang, S. (2024). How connected is withholding capacity to electricity, fossil fuel and carbon markets? perspectives from a high renewable energy consumption economy. *Energy Policy*, 185:113937.

McAfee, R. P. and McMillan, J. (1992). Bidding rings. *The American Economic Review*, 82(3).

Moreau, F. and Panon, L. (2022). Macroeconomic effects of market structure distortions: Evidence from french cartels. *IMF Working Papers*, 2022(104):1.

OECD (2009). *Guidelines for Fighting Bid Rigging in Public Procurement.*

OECD (2022a). Competition in energy markets. *OECD Roundtables on Competition Policy Papers*, (290).

OECD (2022b). Data screening tools for competition investigations. *OECD Roundtables on Competition Policy Papers*, (284).

OECD (2023). Algorithmic competition. *OECD Roundtables on Competition Policy Papers*, (296).

OECD (2025). Oecd guidelines for fighting bid rigging in public procurement. *OECD Publishing*.

Parliament, E. (2026). Internal energy market.

Porter, R. H. (2005). Detecting collusion. *Review of Industrial Organization*, 26(2):147–167.

Porter, R. H. and Zona, J. D. (1993). Detection of bid rigging in procurement auctions. *Journal of political economy*, 101(3):518–538.

Rodríguez, M. J. G., Rodríguez-Montequín, V., Ballesteros-Pérez, P., Love, P. E., and Signor, R. (2022). Collusion detection in public procurement auctions with machine learning algorithms. *Automation in Construction*, 133:104047.

Silveira, D., Vasconcelos, S., Resende, M., and Cajueiro, D. O. (2022). Won't get fooled again: A supervised machine learning approach for screening gasoline cartels. *Energy Economics*, 105:105711.

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 58(1):267–288.

Wallimann, H., Imhof, D., and Huber, M. (2022). A machine learning approach for flagging incomplete bid-rigging cartels. *Computational Economics*, 62(4):1669–1720.

Wallimann, H. and Sticher, S. (2023). On suspicious tracks: Machine-learning based approaches to detect cartels in railway-infrastructure procurement. *Transport Policy*, 143:121–131.

Wolak, F. A. (2003). Measuring unilateral market power in wholesale electricity markets: The california market, 1998–2000. *American Economic Review*, 93(2):425–430.

Zhou, Z.-H. (2012). *Ensemble Methods: Foundations and Algorithms*. Chapman and Hall/CRC.

A — Screen statistics

Table 13: MSD and MGP tenders

Market	Cases	Screens	MW stat.	MW pval.	KS stat.	KS pval.	Mean diff
Narrow	CSUD	MGP offers	287.500	0.678	0.236	0.555	-0.535
Narrow	CSUD	MGP quantity	430.000	0.001	0.574	0.002	-812.184
Narrow	CSUD	MGP accepted offers	262.500	0.943	0.167	0.893	0.559
Narrow	CSUD	MGP accepted quantity	319.000	0.293	0.274	0.370	-103.926
Narrow	CSUD	Variance	11.500	0.000	0.976	0.000	132093.206
Narrow	CSUD	CV	11.500	0.000	0.976	0.000	0.523
Narrow	CSUD	Spread	11.000	0.000	0.976	0.000	686.899
Narrow	CSUD	Kurtosis	57.500	0.000	0.704	0.000	1.750
Narrow	CSUD	Diff.	142.000	0.012	0.501	0.009	8.969
Narrow	CSUD	Diff p.	140.500	0.011	0.512	0.006	0.053
Narrow	CSUD	RD	274.500	0.879	0.334	0.175	-0.118
Narrow	CSUD	RD norm.	270.500	0.943	0.334	0.175	-0.212
Narrow	CSUD	RD alt.	261.000	0.919	0.362	0.114	-0.242
Narrow	CSUD	Skew	45.500	0.000	0.752	0.000	1.197
Narrow	CSUD	KS	33.500	0.000	0.801	0.000	0.270
Narrow	CSUD	Num. bids	266.500	1.000	0.000	1.000	0.000
Narrow	BRNN	MGP offers	18918.500	0.000	0.309	0.000	1.070
Narrow	BRNN	MGP quantity	35234.000	0.000	0.465	0.000	-235.060
Narrow	BRNN	MGP accepted offers	12615.500	0.000	0.400	0.000	1.613
Narrow	BRNN	MGP accepted quantity	18558.500	0.000	0.252	0.000	140.225
Narrow	BRNN	Variance	43032.500	0.000	0.726	0.000	44779.009
Narrow	BRNN	CV	32759.500	0.000	0.548	0.000	0.023
Narrow	BRNN	Spread	44200.000	0.000	0.739	0.000	-152.005
Narrow	BRNN	Kurtosis	15370.500	0.000	0.435	0.000	0.300
Narrow	BRNN	Diff.	52222.500	0.000	0.926	0.000	-226.340
Narrow	BRNN	Diff p.	51785.000	0.000	0.930	0.000	-3.024
Narrow	BRNN	RD	nan	nan	0.465	0.000	-1.549
Narrow	BRNN	RD norm.	nan	nan	0.465	0.000	-2.648
Narrow	BRNN	RD alt.	nan	nan	0.474	0.000	16.093
Narrow	BRNN	Skew	15556.500	0.000	0.470	0.000	1.605
Narrow	BRNN	KS	14783.500	0.000	0.422	0.000	0.108
Narrow	BRNN	Num. bids	26450.000	1.000	0.000	1.000	0.000
Narrow	Combined	MGP offers	28636.000	0.010	0.204	0.000	-8.114

Continued on next page

Market	Cases	Screens	MW stat.	MW pval.	KS stat.	KS pval.	Mean diff
Narrow	Combined	MGP quantity	45094.000	0.000	0.449	0.000	-1698.084
Narrow	Combined	MGP accepted offers	22308.000	0.000	0.294	0.000	-5.741
Narrow	Combined	MGP accepted quantity	28085.500	0.004	0.196	0.000	13.761
Narrow	Combined	Variance	49609.000	0.000	0.601	0.000	50018.923
Narrow	Combined	CV	37916.000	0.003	0.452	0.000	0.059
Narrow	Combined	Spread	51117.000	0.000	0.610	0.000	-89.982
Narrow	Combined	Kurtosis	17987.000	0.000	0.422	0.000	0.449
Narrow	Combined	Diff.	57705.500	0.000	0.784	0.000	-189.373
Narrow	Combined	Diff p.	56028.500	0.000	0.785	0.000	-2.522
Narrow	Combined	RD	nan	nan	0.352	0.000	-0.961
Narrow	Combined	RD norm.	nan	nan	0.352	0.000	-1.748
Narrow	Combined	RD alt.	nan	nan	0.362	0.000	16.112
Narrow	Combined	Skew	20669.000	0.000	0.444	0.000	1.355
Narrow	Combined	KS	16732.000	0.000	0.424	0.000	0.126
Narrow	Combined	Num. bids	36146.500	0.000	0.098	0.158	-0.098
Broad	CSUD	MGP offers	20660.000	0.000	0.330	0.000	7.088
Broad	CSUD	MGP quantity	24107.000	0.000	0.325	0.000	517.787
Broad	CSUD	MGP accepted offers	20496.500	0.000	0.330	0.000	6.081
Broad	CSUD	MGP accepted quantity	9665.000	0.000	0.643	0.000	960.897
Broad	CSUD	Variance	2406.000	0.000	0.939	0.000	38585.216
Broad	CSUD	CV	2638.000	0.000	0.932	0.000	0.311
Broad	CSUD	Spread	2238.000	0.000	0.955	0.000	681.208
Broad	CSUD	Kurtosis	3053.000	0.000	0.955	0.000	8.964
Broad	CSUD	Diff.	54025.000	0.000	0.538	0.000	-8.422
Broad	CSUD	Diff p.	54305.000	0.000	0.538	0.000	-0.128
Broad	CSUD	RD	57914.000	0.000	0.549	0.000	-0.095
Broad	CSUD	RD norm.	58586.500	0.000	0.591	0.000	-0.713
Broad	CSUD	RD alt.	58444.500	0.000	0.561	0.000	-0.710
Broad	CSUD	Skew	3063.000	0.000	0.955	0.000	2.223
Broad	CSUD	KS	3150.000	0.000	0.955	0.000	0.342
Broad	CSUD	Num. bids	7656.000	0.000	0.780	0.000	0.780
Broad	BRNN	MGP offers	21788.500	0.000	0.294	0.000	0.959
Broad	BRNN	MGP quantity	37989.000	0.000	0.453	0.000	-165.538
Broad	BRNN	MGP accepted offers	16719.000	0.000	0.339	0.000	1.355
Broad	BRNN	MGP accepted quantity	23164.500	0.000	0.216	0.000	97.820
Broad	BRNN	Variance	51760.500	0.000	0.800	0.000	29827.005
Broad	BRNN	CV	41352.500	0.000	0.600	0.000	-0.106

Continued on next page

Market	Cases	Screens	MW stat.	MW pval.	KS stat.	KS pval.	Mean diff
Broad	BRNN	Spread	49666.500	0.000	0.727	0.000	-129.727
Broad	BRNN	Kurtosis	16899.500	0.000	0.486	0.000	1.851
Broad	BRNN	Diff.	23210.500	0.000	0.216	0.000	1.363
Broad	BRNN	Diff p.	23099.500	0.000	0.216	0.000	0.020
Broad	BRNN	RD	22971.500	0.000	0.237	0.000	1.256
Broad	BRNN	RD norm.	22947.500	0.000	0.245	0.000	2.378
Broad	BRNN	RD alt.	22940.000	0.000	0.245	0.000	2.851
Broad	BRNN	Skew	20136.500	0.000	0.445	0.000	1.078
Broad	BRNN	KS	13617.000	0.000	0.612	0.000	0.185
Broad	BRNN	Num. bids	30012.500	1.000	0.000	1.000	0.000
Broad	Combined	MGP offers	107128.500	0.000	0.169	0.000	3.650
Broad	Combined	MGP quantity	126776.000	0.749	0.221	0.000	115.179
Broad	Combined	MGP accepted offers	101895.500	0.000	0.169	0.000	3.416
Broad	Combined	MGP accepted quantity	96664.500	0.000	0.330	0.000	535.679
Broad	Combined	Variance	125953.500	0.444	0.332	0.000	34369.574
Broad	Combined	CV	107920.500	0.000	0.255	0.000	0.110
Broad	Combined	Spread	106849.500	0.000	0.530	0.000	290.876
Broad	Combined	Kurtosis	49015.500	0.000	0.717	0.000	5.540
Broad	Combined	Diff.	156552.000	0.000	0.202	0.000	-3.712
Broad	Combined	Diff p.	156230.500	0.000	0.202	0.000	-0.057
Broad	Combined	RD	156296.500	0.000	0.228	0.000	0.555
Broad	Combined	RD norm.	159420.000	0.000	0.255	0.000	0.775
Broad	Combined	RD alt.	158807.500	0.000	0.240	0.000	1.004
Broad	Combined	Skew	54273.500	0.000	0.705	0.000	1.672
Broad	Combined	KS	19946.000	0.000	0.778	0.000	0.266
Broad	Combined	Num. bids	102348.500	0.000	0.405	0.000	0.405

B — Ensemble method performance

Table 14: Ensemble learner performance measures

Market	Case	Model	Screens	Accuracy	Recall	Specificity	Precision	F1 score	AUC	True Positive	False Positive	True Negative	False Negative	RFP	False discovery rate	Missing rate
Narrow	CSUD	Normal	Orginial	96.429	97.231	95.000	98.182	97.595	99.379	10.900	0.200	2.600	0.300	0.050	1.802	2.679
Narrow	CSUD	Demeaned	Orginial	97.143	97.231	97.500	99.091	98.071	99.379	10.900	0.100	2.700	0.300	0.025	0.909	2.679
Narrow	CSUD	Std.	Orginial	95.714	97.231	92.500	97.273	97.119	99.379	10.900	0.300	2.500	0.300	0.075	2.679	2.679
Narrow	CSUD	Normal	With MGP	97.143	97.231	97.500	99.091	98.071	99.407	10.900	0.100	2.700	0.300	0.025	0.909	2.679
Narrow	CSUD	Demeaned	With MGP	96.429	97.231	95.000	98.182	97.595	99.407	10.900	0.200	2.600	0.300	0.050	1.802	2.679
Narrow	CSUD	Std.	With MGP	97.143	97.231	97.500	99.091	98.071	99.407	10.900	0.100	2.700	0.300	0.025	0.909	2.679
Broad	CSUD	Normal	Orginial	99.545	99.265	99.865	99.831	99.543	99.994	64.500	0.100	66.900	0.500	0.001	0.155	0.769
Broad	CSUD	Demeaned	Orginial	99.545	99.569	99.544	99.545	99.554	99.994	64.700	0.300	66.700	0.300	0.005	0.462	0.462
Broad	CSUD	Std.	Orginial	99.848	99.851	99.865	99.831	99.839	99.994	64.900	0.100	66.900	0.100	0.001	0.154	0.154
Broad	CSUD	Normal	With MGP	99.697	99.569	99.865	99.831	99.697	99.998	64.700	0.100	66.900	0.300	0.001	0.154	0.462
Broad	CSUD	Demeaned	With MGP	99.545	99.408	99.711	99.683	99.542	99.988	64.600	0.200	66.800	0.400	0.003	0.309	0.615
Broad	CSUD	Std.	With MGP	99.848	99.851	99.865	99.831	99.839	99.994	64.900	0.100	66.900	0.100	0.001	0.154	0.154
Narrow	BRNN	Normal	Orginial	96.696	98.418	94.905	95.237	96.763	99.716	57.500	2.900	53.700	0.900	0.051	4.801	1.541
Narrow	BRNN	Demeaned	Orginial	96.783	98.418	95.087	95.398	96.842	99.716	57.500	2.800	53.800	0.900	0.049	4.643	1.541
Narrow	BRNN	Std.	Orginial	96.783	98.584	94.905	95.245	96.849	99.748	57.600	2.900	53.700	0.800	0.051	4.793	1.370
Narrow	BRNN	Normal	With MGP	97.304	98.792	95.786	96.039	97.355	99.766	57.700	2.400	54.200	0.700	0.042	3.993	1.199
Narrow	BRNN	Demeaned	With MGP	97.304	98.625	95.958	96.192	97.352	99.772	57.600	2.300	54.300	0.800	0.040	3.840	1.370
Narrow	BRNN	Std.	With MGP	97.217	98.769	95.614	95.877	97.265	99.739	57.700	2.500	54.100	0.700	0.044	4.153	1.199
Broad	BRNN	Normal	Orginial	96.992	98.658	95.289	95.473	97.001	99.459	60.800	2.900	58.500	0.800	0.047	4.553	1.299
Broad	BRNN	Demeaned	Orginial	97.317	98.992	95.631	95.797	97.324	99.524	61.000	2.700	58.700	0.600	0.044	4.239	0.974
Broad	BRNN	Std.	Orginial	97.073	98.500	95.652	95.746	97.062	99.807	60.700	2.700	58.700	0.900	0.043	4.259	1.461
Broad	BRNN	Normal	With MGP	97.236	98.992	95.453	95.606	97.245	99.493	61.000	2.800	58.600	0.600	0.045	4.389	0.974
Broad	BRNN	Demeaned	With MGP	97.642	99.327	95.979	96.081	97.642	99.554	61.200	2.500	58.900	0.400	0.040	3.925	0.649
Broad	BRNN	Std.	With MGP	97.154	98.658	95.622	95.785	97.156	99.849	60.800	2.700	58.700	0.800	0.044	4.252	1.299
Narrow	Combined	Normal	Orginial	95.581	97.122	93.792	94.830	95.924	99.225	67.900	3.700	55.400	2.000	0.062	5.168	2.861
Narrow	Combined	Demeaned	Orginial	95.736	97.551	93.660	94.697	96.060	99.264	68.200	3.800	55.300	1.700	0.063	5.278	2.432
Narrow	Combined	Std.	Orginial	95.349	96.690	93.797	94.829	95.693	99.416	67.600	3.700	55.400	2.300	0.062	5.189	3.290
Narrow	Combined	Normal	With MGP	96.202	97.988	94.150	95.098	96.492	99.598	68.500	3.500	55.600	1.400	0.058	4.861	2.003
Narrow	Combined	Demeaned	With MGP	96.202	97.691	94.483	95.342	96.474	99.486	68.300	3.300	55.800	1.600	0.055	4.609	2.289
Narrow	Combined	Std.	With MGP	96.279	97.835	94.468	95.399	96.559	99.624	68.400	3.300	55.800	1.500	0.055	4.603	2.146
Broad	Combined	Normal	Orginial	97.804	97.512	98.121	98.121	97.804	99.571	123.900	2.400	125.500	3.200	0.019	1.900	2.518
Broad	Combined	Demeaned	Orginial	97.961	98.142	97.806	97.822	97.973	99.645	124.700	2.800	125.100	2.400	0.022	2.196	1.888
Broad	Combined	Std.	Orginial	98.392	98.895	97.893	97.902	98.391	99.892	125.700	2.700	125.200	1.400	0.021	2.103	1.101
Broad	Combined	Normal	With MGP	98.118	97.980	98.271	98.280	98.124	99.604	124.500	2.200	125.700	2.600	0.017	1.736	2.046
Broad	Combined	Demeaned	With MGP	97.961	97.975	97.961	97.980	97.968	99.683	124.500	2.600	125.300	2.600	0.020	2.046	2.046
Broad	Combined	Std.	With MGP	98.431	98.745	98.125	98.138	98.432	99.901	125.500	2.400	125.500	1.600	0.019	1.876	1.259

C — Ensemble method weights

Table 15: Ensemble learner weights

Case	Market	Screens	Model	Random Forests	Bagged Decision Trees	Lasso Logit	SVM	Logit
CSUD	Narrow	Original	Normal	0.221	0.221	0.121	0.221	0.215
CSUD	Narrow	Original	Demeaned	0.201	0.201	0.204	0.203	0.191
CSUD	Narrow	Original	Std.	0.204	0.204	0.197	0.199	0.196
CSUD	Narrow	With MGP	Normal	0.205	0.204	0.180	0.205	0.206
CSUD	Narrow	With MGP	Demeaned	0.205	0.204	0.180	0.207	0.204
CSUD	Narrow	With MGP	Std.	0.204	0.202	0.196	0.200	0.198
CSUD	Broad	Original	Normal	0.207	0.206	0.183	0.202	0.201
CSUD	Broad	Original	Demeaned	0.208	0.208	0.175	0.204	0.205
CSUD	Broad	Original	Std.	0.202	0.202	0.198	0.197	0.201
CSUD	Broad	With MGP	Normal	0.206	0.206	0.185	0.202	0.201
CSUD	Broad	With MGP	Demeaned	0.208	0.208	0.177	0.204	0.204
CSUD	Broad	With MGP	Std.	0.202	0.202	0.198	0.198	0.200
BRNN	Narrow	Original	Normal	0.242	0.241	0.135	0.143	0.240
BRNN	Narrow	Original	Demeaned	0.231	0.231	0.169	0.139	0.230
BRNN	Narrow	Original	Std.	0.201	0.201	0.199	0.199	0.199
BRNN	Narrow	With MGP	Normal	0.241	0.242	0.136	0.142	0.240
BRNN	Narrow	With MGP	Demeaned	0.232	0.232	0.167	0.139	0.231
BRNN	Narrow	With MGP	Std.	0.201	0.202	0.200	0.198	0.199
BRNN	Broad	Original	Normal	0.213	0.214	0.158	0.204	0.212
BRNN	Broad	Original	Demeaned	0.202	0.203	0.200	0.194	0.201
BRNN	Broad	Original	Std.	0.201	0.202	0.201	0.197	0.200
BRNN	Broad	With MGP	Normal	0.207	0.208	0.179	0.199	0.207
BRNN	Broad	With MGP	Demeaned	0.202	0.203	0.199	0.194	0.202
BRNN	Broad	With MGP	Std.	0.201	0.202	0.201	0.197	0.200
Combined	Narrow	Original	Normal	0.240	0.240	0.145	0.138	0.236
Combined	Narrow	Original	Demeaned	0.240	0.241	0.145	0.137	0.238
Combined	Narrow	Original	Std.	0.201	0.202	0.199	0.200	0.198
Combined	Narrow	With MGP	Normal	0.239	0.237	0.147	0.141	0.236
Combined	Narrow	With MGP	Demeaned	0.241	0.240	0.139	0.140	0.239
Combined	Narrow	With MGP	Std.	0.202	0.201	0.199	0.200	0.199
Combined	Broad	Original	Normal	0.210	0.209	0.171	0.205	0.205
Combined	Broad	Original	Demeaned	0.207	0.207	0.180	0.202	0.203
Combined	Broad	Original	Std.	0.202	0.202	0.199	0.198	0.200
Combined	Broad	With MGP	Normal	0.206	0.206	0.184	0.202	0.202
Combined	Broad	With MGP	Demeaned	0.206	0.206	0.183	0.201	0.203
Combined	Broad	With MGP	Std.	0.202	0.202	0.199	0.198	0.199

D — Benchmark model performance

D.1. SVM model performance

Table 16: SVM model performance

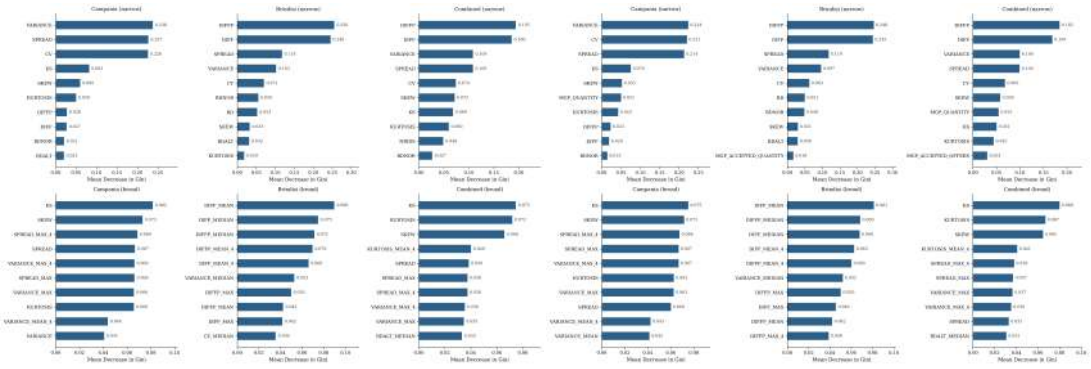
Case	Market	Screens	Model	Accuracy	Recall	Specificity	RFP
CSUD	Narrow	Orginial	Normal	0.979	0.972	1.000	0.000
CSUD	Narrow	Orginial	Demeaned	0.979	0.972	1.000	0.000
CSUD	Narrow	Orginial	Std.	0.950	0.972	0.900	0.100
CSUD	Narrow	With MGP	Normal	0.979	0.972	1.000	0.000
CSUD	Narrow	With MGP	Demeaned	0.979	0.972	1.000	0.000
CSUD	Narrow	With MGP	Std.	0.964	0.972	0.950	0.050
CSUD	Broad	Orginial	Normal	0.983	0.965	1.000	0.000
CSUD	Broad	Orginial	Demeaned	0.983	0.965	1.000	0.000
CSUD	Broad	Orginial	Std.	0.979	0.965	0.993	0.007
CSUD	Broad	With MGP	Normal	0.983	0.965	1.000	0.000
CSUD	Broad	With MGP	Demeaned	0.983	0.965	1.000	0.000
CSUD	Broad	With MGP	Std.	0.980	0.965	0.994	0.006
BRNN	Narrow	Orginial	Normal	0.557	0.269	0.860	0.140
BRNN	Narrow	Orginial	Demeaned	0.569	0.296	0.858	0.142
BRNN	Narrow	Orginial	Std.	0.959	0.975	0.942	0.058
BRNN	Narrow	With MGP	Normal	0.557	0.269	0.860	0.140
BRNN	Narrow	With MGP	Demeaned	0.570	0.298	0.858	0.142
BRNN	Narrow	With MGP	Std.	0.962	0.980	0.942	0.058
BRNN	Broad	Orginial	Normal	0.933	0.993	0.876	0.124
BRNN	Broad	Orginial	Demeaned	0.934	0.995	0.875	0.125
BRNN	Broad	Orginial	Std.	0.951	0.988	0.913	0.087
BRNN	Broad	With MGP	Normal	0.933	0.993	0.876	0.124
BRNN	Broad	With MGP	Demeaned	0.934	0.995	0.875	0.125
BRNN	Broad	With MGP	Std.	0.951	0.990	0.911	0.089
Combined	Narrow	Orginial	Normal	0.522	0.608	0.470	0.530
Combined	Narrow	Orginial	Demeaned	0.570	0.605	0.578	0.422
Combined	Narrow	Orginial	Std.	0.954	0.971	0.934	0.066
Combined	Narrow	With MGP	Normal	0.545	0.612	0.517	0.483
Combined	Narrow	With MGP	Demeaned	0.581	0.617	0.586	0.414
Combined	Narrow	With MGP	Std.	0.955	0.975	0.930	0.070
Combined	Broad	Orginial	Normal	0.967	0.967	0.967	0.033
Combined	Broad	Orginial	Demeaned	0.960	0.967	0.953	0.047
Combined	Broad	Orginial	Std.	0.971	0.972	0.970	0.030
Combined	Broad	With MGP	Normal	0.968	0.967	0.970	0.030
Combined	Broad	With MGP	Demeaned	0.960	0.967	0.953	0.047
Combined	Broad	With MGP	Std.	0.971	0.972	0.970	0.030

Table 17: Lasso model performance

Case	Market	Screens	Model	Accuracy	Recall	Specificity	RFP
CSUD	Narrow	Orginial	Normal	0.593	0.511	1.000	0.000
CSUD	Narrow	Orginial	Demeaned	0.979	0.972	1.000	0.000
CSUD	Narrow	Orginial	Std.	0.936	0.972	0.817	0.183
CSUD	Narrow	With MGP	Normal	0.871	0.972	0.400	0.600
CSUD	Narrow	With MGP	Demeaned	0.871	0.972	0.400	0.600
CSUD	Narrow	With MGP	Std.	0.936	0.972	0.817	0.183
CSUD	Broad	Orginial	Normal	0.858	0.870	0.848	0.152
CSUD	Broad	Orginial	Demeaned	0.833	0.855	0.813	0.187
CSUD	Broad	Orginial	Std.	0.986	0.977	0.996	0.004
CSUD	Broad	With MGP	Normal	0.877	0.886	0.868	0.132
CSUD	Broad	With MGP	Demeaned	0.840	0.891	0.790	0.210
CSUD	Broad	With MGP	Std.	0.986	0.977	0.994	0.006
BRNN	Narrow	Orginial	Normal	0.550	1.000	0.087	0.913
BRNN	Narrow	Orginial	Demeaned	0.778	1.000	0.556	0.444
BRNN	Narrow	Orginial	Std.	0.958	0.979	0.937	0.063
BRNN	Narrow	With MGP	Normal	0.560	1.000	0.109	0.891
BRNN	Narrow	With MGP	Demeaned	0.768	0.720	0.856	0.144
BRNN	Narrow	With MGP	Std.	0.968	0.988	0.947	0.053
BRNN	Broad	Orginial	Normal	0.720	0.988	0.447	0.553
BRNN	Broad	Orginial	Demeaned	0.958	0.998	0.916	0.084
BRNN	Broad	Orginial	Std.	0.969	0.980	0.958	0.042
BRNN	Broad	With MGP	Normal	0.833	0.988	0.677	0.323
BRNN	Broad	With MGP	Demeaned	0.958	0.998	0.916	0.084
BRNN	Broad	With MGP	Std.	0.968	0.980	0.957	0.043
Combined	Narrow	Orginial	Normal	0.580	0.988	0.098	0.902
Combined	Narrow	Orginial	Demeaned	0.587	0.988	0.110	0.890
Combined	Narrow	Orginial	Std.	0.947	0.971	0.917	0.083
Combined	Narrow	With MGP	Normal	0.607	0.978	0.174	0.826
Combined	Narrow	With MGP	Demeaned	0.537	0.201	0.947	0.053
Combined	Narrow	With MGP	Std.	0.953	0.975	0.926	0.074
Combined	Broad	Orginial	Normal	0.807	0.974	0.644	0.356
Combined	Broad	Orginial	Demeaned	0.855	0.979	0.731	0.269
Combined	Broad	Orginial	Std.	0.975	0.972	0.979	0.021
Combined	Broad	With MGP	Normal	0.889	0.974	0.806	0.194
Combined	Broad	With MGP	Demeaned	0.876	0.978	0.775	0.225
Combined	Broad	With MGP	Std.	0.973	0.970	0.975	0.025

E — Variable relevance

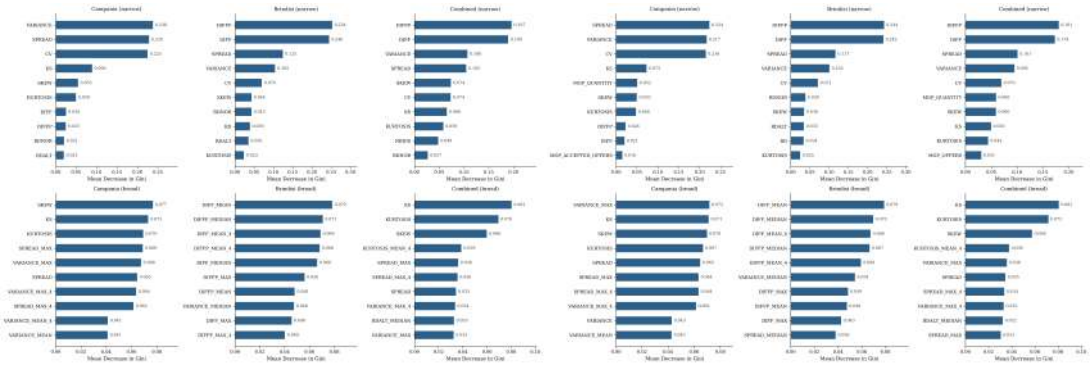
Figure 7: Variable relevance: normal models



(a) Normal screens

(b) Normal screens with MGP screens

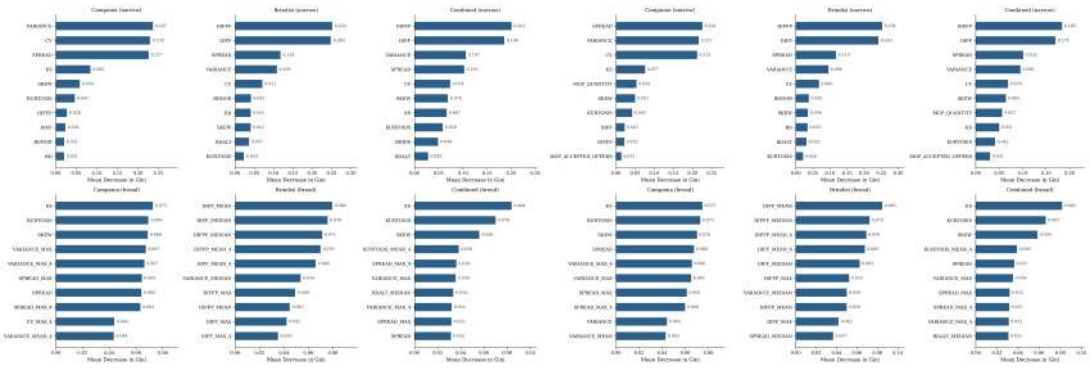
Figure 8: Variable relevance: demeaned models



(a) Demeaned screens

(b) Demeaned screens with MGP screens

Figure 9: Variable relevance: standardized models



(a) Standardized screens

(b) Standardized screens with MGP screens